

1 Authoring Guidelines

To support practitioners in writing UCSs for IoT systems, we define lightweight authoring guidelines. The purpose of these guidelines is not to replace RUCM, but to help practitioners write use case specifications that are sufficiently structured for automated extraction while remaining easy to adopt in practice.

The guidelines are as follows:

- **Identify the actor or IoT node performing each action.** Each step should clearly specify the component responsible for the action, such as Fitbit, SmartPhone, CloudApp, or Buddy.
- **Use explicit action keywords.** Practitioners should use clear verbs such as SENDS, RECEIVES, CHECKS THAT, SETS, MOVES, or ROTATES to indicate the operation performed in each step.
- **Distinguish interaction steps from internal processing steps.** Interactions between two nodes should explicitly identify the sender, receiver, exchanged data or command, and communication protocol. Internal processing steps should describe the computation or state change performed within a single node.
- **Specify target nodes and communication protocols.** When a step involves communication between nodes, the target node should be indicated using TO or FROM, and the communication protocol should be specified using OVER, such as OVER HTTP or OVER WebSocket.
- **Express inputs, conditions, and thresholds using measurable values.** Inputs and constraints should be written explicitly, for example, `heartrate=80 bpm`, `distance=1`, `speed=1`, or `heartrate is between 60 and 100 bpm`.
- **Separate the main and alternative flows.** Practitioners should clearly distinguish the basic flow from bounded alternative flows and specific alternative flows. Conditional behavior should be written using IF, THEN, and ENDIF.
- **Define postconditions for each flow.** Each flow should include an expected outcome that can be used as an oracle during test generation and execution, such as `Execution_status is WHEEL_MOVE_FINISHED` or `Execution_status is NOK`.

These guidelines help practitioners write UCSs that remain close to RUCM while making IoT-specific information explicit enough for FUNEETIS to extract scenarios, actions, inputs, target information, constraints, and expected outcomes.

2 Example

The following step illustrates an interaction statement between two IoT nodes:

```
CloudApp SENDS MOVE command TO Buddy WITH distance=1 and speed=1  
OVER WebSocket
```

In this step, **CloudApp** is the sender node, **Buddy** is the target node, **SENDS** defines the interaction action, **distance=1** and **speed=1** represent measurable input parameters, and **WebSocket** specifies the communication protocol. In contrast, the following statement represents an internal processing action within a single node:

```
CloudApp CHECKS THAT heartrate is between 60 and 100 bpm
```

Finally, the following statement specifies an expected execution outcome used as a postcondition:

```
Execution_status is WHEEL_MOVE_FINISHED
```